PHRÓNĒSIS ACADEMIC HUB

CHM 102: Organic chemistry

1. Define stereochemistry and explain its importance in organic chemistry.

2. Distinguish between constitutional (structural) isomers and stereoisomers with suitable examples.

3. Explain the concept of stereoisomerism.

4. Describe the different classes of stereoisomers.

5. What is chirality? State the conditions necessary for a molecule to be chiral.

6. Define an asymmetric (chiral) carbon atom and explain how it gives rise to optical activity.

7. Differentiate between enantiomers and diastereomers.

8. Explain the concept of optical isomerism.

9. Describe the properties of enantiomers.

10. What are meso compounds? Explain why they are optically inactive.

11. State and explain the significance of plane of symmetry in stereochemistry.

12. Explain the difference between racemic mixtures and optically active compounds.

13. Discuss the methods of representing three-dimensional structures of organic compounds.

14. Explain Fischer projection formula and state its rules.

15. Explain Newman projection formula and state its applications.

16. Describe Sawhorse projection and explain its significance.

17. Differentiate between eclipsed and staggered conformations of ethane.

18. Explain conformational isomerism using ethane as an example.

19. Discuss the conformations of cyclohexane.

20. Compare chair, boat and twist-boat conformations of cyclohexane.

21. Explain axial and equatorial bonds in cyclohexane.

22. Discuss ring flipping in cyclohexane and its significance.

23. Explain cis-trans (geometrical) isomerism.

24. State the conditions necessary for geometrical isomerism.
25. Differentiate between cis-trans nomenclature and E/Z nomenclature.
26. Explain the Cahn-Ingold-Prelog (CIP) priority rules.
27. Assign R/S configuration to chiral compounds (theory).
28. Discuss the pharmaceutical importance of stereochemistry.
29. Explain the effect of stereochemistry on biological activity of drugs.
30. Write short notes on:
    1. Chirality
    2. Optical activity
    3. Fischer projection
    4. Conformational isomerism
    5. Geometrical isomerism